Unit 6.01: Permutations & Combinations

Q1. Find the value of the integer n ge 3 that satisfies the equation {}^{n}mathrm{P}_{3} = 10 times {}^{n}mathrm{C}_{2}

[x] A: n = 7
[ ] B: n = 12
[ ] C: n = 5
[ ] D: n = 8

Solution: Using the algebraic definitions of permutations and combinations: {}^{n}mathrm{P}_{3} = frac{n!}{n - 3!} = nn - 1n - 2 {}^{n}mathrm{C}_{2} = frac{n!}{2!n - 2!} = frac{nn - 1}{2} Substitute these into the given equation: nn - 1n - 2 = 10 times frac{nn - 1}{2} nn - 1n - 2 = 5nn - 1 Since n ge 3 , we have nn - 1 ne 0 . Dividing both sides by nn - 1 gives: n - 2 = 5 implies n = 7 Common misconceptions: - Forgetting the 2! denominator in {}^{n}mathrm{C}_{2} treating it as {}^{n}mathrm{P}_{2} gives n - 2 = 10 implies n = 12 . - Dividing by nn - 1 and erroneously writing n = 5 instead of n - 2 = 5 gives n = 5 . - Confusing the factorial simplification to set n - 3 = 5 leads to n = 8 . [M1][A1]

Q2. A committee of 5 members is to be selected from a group consisting of n men and 6 women, where n ge 4 . Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n .

[x] A: 13
[ ] B: 10
[ ] C: 8
[ ] D: 7

Solution: To form a committee of 5 members: 1. **Committees with exactly 3 men:** Since there are 5 members in total, the committee must contain 3 men and 5 - 3 = 2 women. text{Number of ways} = binom{n}{3} times binom{6}{2} = binom{n}{3} times frac{6 times 5}{2} = 15binom{n}{3} 2. **Committees with exactly 4 men:** The committee must contain 4 men and 5 - 4 = 1 woman. text{Number of ways} = binom{n}{4} times binom{6}{1} = 6binom{n}{4} 3. **Equating the two expressions:** 15binom{n}{3} = 6binom{n}{4} Divide both sides by 3: 5binom{n}{3} = 2binom{n}{4} Using the relation binom{n}{4} = frac{n - 3}{4}binom{n}{3} : 5binom{n}{3} = 2 times frac{n - 3}{4}binom{n}{3} = frac{n - 3}{2}binom{n}{3} Since n ge 4 , binom{n}{3} neq 0 , so we can divide both sides by binom{n}{3} : 5 = frac{n - 3}{2} implies 10 = n - 3 implies n = 13 **Common Traps:** - Setting binom{n}{3} = binom{n}{4} ignores the selection of the women needed to complete the 5-member committee, giving frac{n-3}{4} = 1 implies n = 7 . - Forgetting to add 3 after solving n - 3 = 10 leads to the distractor 10. [M1][A1]

Q3. Six-digit numbers are formed using digits chosen from the set {1, 2, 3, 4, 5, 6, 7, 8}, with no digit repeated in any number. Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.

[x] A: 8160
[ ] B: 7680
[ ] C: 9120
[ ] D: 2880

Solution: We use complementary counting: text{Number of valid arrangements} = text{Total odd 6-digit numbers} - text{Odd 6-digit numbers where } 2 text{ and } 4 text{ are adjacent}. **Step 1: Total number of 6-digit odd numbers** - For the number to be odd, the last digit units place must be chosen from the odd digits {1, 3, 5, 7}, giving 4 choices. - The remaining 5 positions are filled by selecting and arranging 5 digits from the remaining 7 digits in the set: P7, 5 = 7 times 6 times 5 times 4 times 3 = 2520 - Total odd 6-digit numbers = 4 times 2520 = 10080. **Step 2: Number of 6-digit odd numbers where 2 and 4 are adjacent** - Treat the pair of digits 2 and 4 as a single block B, which can be ordered as 2, 4 or 4, 2 2 internal arrangements. - Since the number must be odd, the units digit must be odd 4 choices from {1, 3, 5, 7}. Because both 2 and 4 are even, the block B cannot occupy the final position. - Thus, the block B must occupy two adjacent positions entirely within the first 5 digits. The possible adjacent pairs for B are positions 1, 2, 2, 3, 3, 4, and 4, 5, giving 4 possible positions. - The remaining 3 unfilled positions among the first 5 places are filled from the remaining 8 - 1 - 2 = 5 available digits: P5, 3 = 5 times 4 times 3 = 60 - Number of odd numbers with 2 and 4

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Q4. Given that {}^{n}mathrm{P}_{3} = 8,{}^{n}mathrm{C}_{2}, find the value of the integer n , where n ge 3 .

[x] A: 6

[ ] B: 8

[ ] C: 10

[ ] D: 12

Solution: By definition of permutations and combinations: {}^{n}mathrm{P}_{3} = frac{n!}{n-3!} = nn-1n-2 and {}^{n}mathrm{C}_{2} = frac{n!}{2!n-2!} = frac{nn-1}{2}. Substitute these expressions into the given equation: nn-1n-2 = 8 leftfrac{nn-1}{2}right = 4nn-1. Since n ge 3 , nn-1 neq 0 . Dividing both sides by nn-1 gives: n - 2 = 4 implies n = 6. Common error: Omitting the factor of 2! in the denominator of {}^{n}mathrm{C}_{2} leads to n - 2 = 8 implies n = 10 . [M1][A1]

Q5. A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600 . Find the value of the integer n .

[ ] A: 6

[x] B: 7

[ ] C: 8

[ ] D: 9

Solution: We can find the number of arrangements where Ali and Priya do not stand next to each other using the complementary method or the insertion gap method. **Method 1: Complementary Counting** 1. The total number of unrestricted arrangements of the n students in a straight line is n! . 2. Consider the arrangements in which Ali and Priya *do* stand next to each other: - Treat Ali and Priya as a single block. There are now n - 1 entities to arrange, which can be done in n - 1! ways. - Within the block, Ali and Priya can arrange themselves in 2! = 2 ways. - So, the number of arrangements where they are together is 2n - 1! . 3. Subtracting this from the total gives the number of arrangements where they are separated: n! - 2n - 1! = nn - 1! - 2n - 1! = n - 2n - 1! **Method 2: Gap Method** 1. Arrange the other n - 2 students first, which gives n - 2! ways. 2. This creates n - 2 + 1 = n - 1 available spaces including the ends. 3. Placing Ali and Priya into two distinct spaces ensures they are not adjacent: {}^{n-1}mathrm{P}_{2} = n - 1n - 2 ways. 4. Total arrangements: n - 2! times n - 1n - 2 = n - 2n - 1! . Equating this to 3600 : n - 2n - 1! = 3600 Testing integer values for n ge 3 : - For n = 6 : 6 - 25! = 4 times 120 = 480 - For n = 7 : 7 - 26! = 5 times 720 = 3600 - For n = 8 : 8 - 27! = 6 times 5040 = 30240 Thus, n = 7 . *Common Misconception Trap:* Forgetting that Ali and Priya can switch places within their unit gives n! - n - 1! = n - 1n - 1! , which fails to produce an integer solution among the options. [M1][A1]

Q6. Two parallel lines, L_1 and L_2 , have n and n + 2 distinct points marked on them respectively, where n is an integer with n ge 2 . Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is 6 : 5 , find the total number of triangles that can be formed.

[x] A: 175

[ ] B: 210

[ ] C: 220

[ ] D: 96

Solution: 1. **Determine the number of triangles, T :** Three points form a triangle if and only if they are not all on the same straight line. - Case 1: 1 point from L_1 and 2 points from L_2 : binom{n}{1} binom{n+2}{2} = n cdot frac{n+2n+1}{2} - Case 2: 2 points from L_1 and 1 point from L_2 : binom{n}{2} binom{n+2}{1} = frac{nn-1}{2} cdot n+2 Summing both cases: T = frac{nn+2}{2} left[n+1 + n-1 right] = frac{nn+2}{2} 2n = n^2n+2 2. **Determine the number of quadrilaterals, Q :** To ensure no three vertices are collinear, exactly 2 points must be chosen from L_1 and 2 points from L_2 : Q = binom{n}{2} binom{n+2}{2} = frac{nn-1}{2} cdot frac{n+2n+1}{2} = frac{nn-1n+1n+2}{4} 3. **Use the given ratio to find n :** frac{Q}{T} = frac{frac{nn-1n+1n+2}{4}}{n^2n+2} = frac{n-1n+1}{4n} = frac{n^2 -

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Q7. Given that {}^ntext{P}_2 = dbinom{n}{3}, where n is an integer such that n ge 3, find the value of n.

[x] A: 8

[ ] B: 5

[ ] C: 3

[ ] D: 6

Solution: Using the algebraic definitions of permutations and combinations: {}^ntext{P}_2 = frac{n!}{n-2!} = nn-1 and binom{n}{3} = frac{n!}{3!n-3!} = frac{nn-1n-2}{3!} = frac{nn-1n-2}{6} Setting them equal gives: nn-1 = frac{nn-1n-2}{6} Since n ge 3, nn-1 ne 0. Dividing both sides by nn-1: 1 = frac{n-2}{6} Multiplying by 6: 6 = n - 2 implies n = 8 Common misconceptions: - Confusing 3! with 3 leads to n - 2 = 3 implies n = 5. - Forgetting the division by 3! entirely treats the combination as a permutation {}^ntext{P}_3, yielding n - 2 = 1 implies n = 3. - Misinterpreting n - 2 = 6 by taking the coefficient directly yields n = 6. [M1][A1]

Q8. A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

[x] A: 60

[ ] B: 120

[ ] C: 55

[ ] D: 36

Solution: binom{5}{2} times binom{4}{2} = frac{5times4}{2} times frac{4times3}{2} = 60 . [M1][A1]